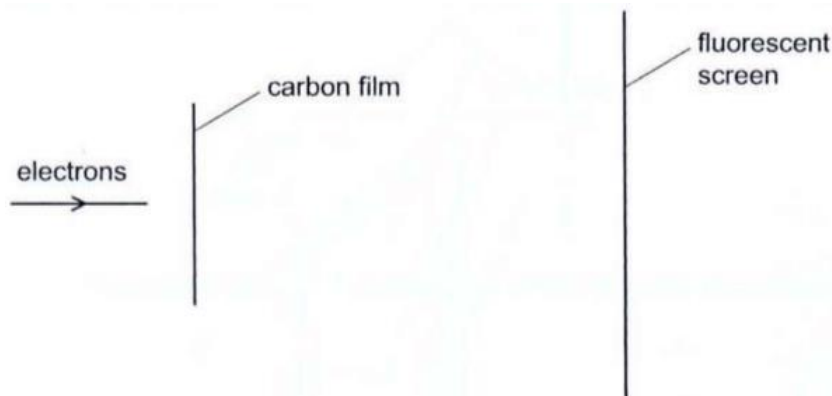


- 28 Electrons with velocity v travel through a vacuum and are incident on a thin carbon film, as shown.



The electrons produce a pattern of concentric circles on the fluorescent screen.

What causes the pattern and which change to the pattern occurs when the velocity v is increased?

	cause	change to pattern
A	diffraction	diameters of circles increase
B	diffraction	diameters of circles decrease
C	refraction	diameters of circles increase
D	refraction	diameters of circles decrease

- | | | |
|-----------|----------|--|
| 28 | B | v increases $\rightarrow p$ increases and by $\lambda = h/p$, λ decreases
by $d \sin \theta = n\lambda$, θ decreases and hence diameter of rings decreases |
|-----------|----------|--|

- 29 The rest mass of the deuterium ${}^2\text{H}$ is equivalent to an energy of 1876 MeV. The rest mass